SMARTBRIDGE Let's Bridge the Gap

SKILL WALLET A SmartBridge Product

Project Initialization and Planning Phase

Date	29 June 2026
Team ID	
Project Title	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed **Smart Lender** system is an AI-powered web application designed to improve the loan approval process using machine learning. It predicts the creditworthiness of loan applicants based on financial and personal information, helping banks and financial institutions make faster, more accurate, and data-driven lending decisions. The solution reduces manual effort, minimizes human error, improves operational efficiency, and enhances customer satisfaction by providing instant loan eligibility predictions.

Project Overview	
Objective	The primary objective of this project is to develop an intelligent loan eligibility prediction system using Machine Learning techniques. The system aims to automate the loan approval process by providing quick, reliable, and accurate predictions based on applicant information while reducing manual effort and improving decision-making.
Scope	The Smart Lender application enables users to register, log in securely, submit loan applications, receive AI-based loan eligibility predictions, view prediction history, access dashboard analytics, and download professional PDF reports. The solution is designed to support banks and financial institutions in making efficient lending decisions.
Problem Statement	
Description	Traditional loan approval processes are time-consuming, manually intensive, and prone to human errors and inconsistent decision-making. Financial institutions often face challenges in evaluating large volumes of loan applications while maintaining accuracy and fairness.
Impact	Automating the loan approval process using machine learning improves prediction accuracy, reduces loan processing time, minimizes financial risks, lowers operational costs, and increases customer satisfaction by delivering faster and more transparent loan decisions.
Proposed Solution	
Approach	The Smart Lender system employs a trained **Random Forest Machine Learning model** to analyze applicant information such as income, employment status, education, loan amount, credit history,

	and property area. After preprocessing the input data, the system predicts whether the loan should be approved or rejected, calculates the confidence score and risk level, stores the prediction in the database, and generates a downloadable PDF report.
Key Features	• Secure User Registration and Login • AI-Based Loan Eligibility Prediction • Random Forest Machine Learning Model • Confidence Score and Risk Assessment • Dashboard with Loan Analytics • Prediction History Management • PDF Report Generation • Responsive Web Interface • SQLite Database Integration • Fast and Accurate Decision Support

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Intel Core i5/Ryzen 5 Processor or higher
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	NumPy, Pandas, ReportLab, Pickle
Development Environment	IDE	Visual Studio Code / PyCharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv UCI dataset, 690, csv
Deployment	Cloud Platform	Render
	Version Control	Git & GitHub